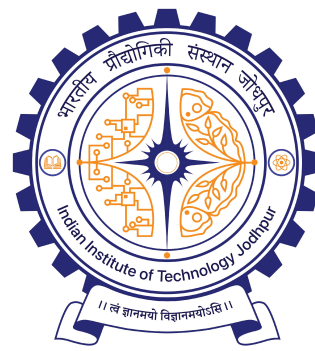




Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-3: 1 Class



*Please Listen carefully
&*

Keep your phone(s) on silent mode

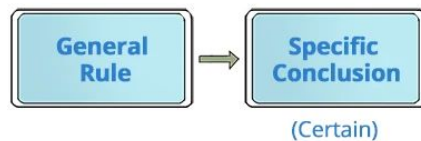
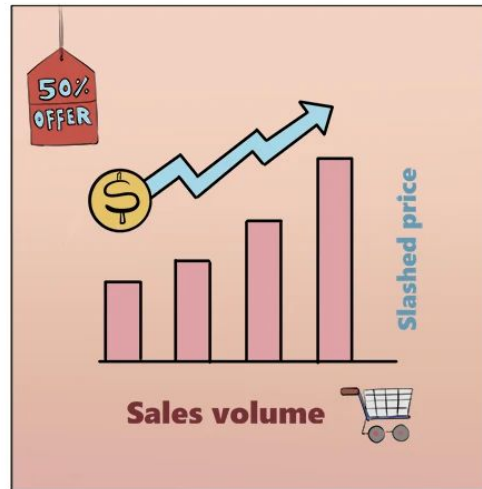
Types of reasoning

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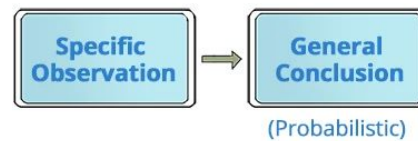
Deductive: General rule □ Specific conclusion (always true)

Inductive: Specific observation □ General conclusion (may be true)

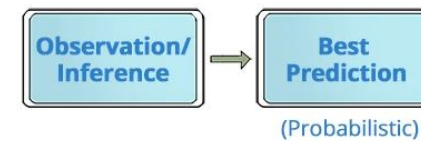
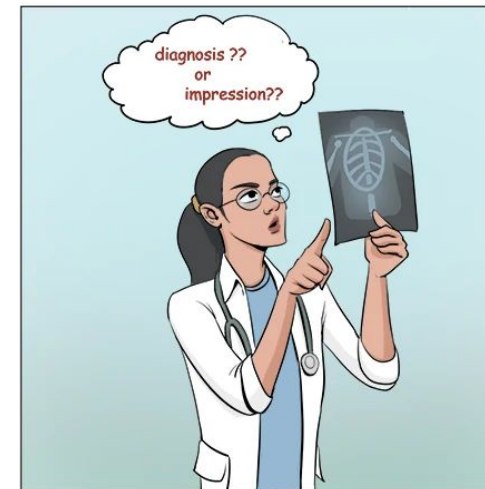
Abductive: Incomplete observation □ best prediction (may be true)



Deductive Reasoning



Inductive Reasoning



Abductive Reasoning

Deductive Reasoning



- Deductive reasoning is a basic form of reasoning, in which it starts out with a general statement, or hypothesis, and examines the possibilities to reach a specific, logical conclusion.
- There is a first premise, then a second premise and finally an inference (a conclusion based on reasoning and evidence).
- A common form of deductive reasoning is the **syllogism**, in which two statements — a **major premise** and a **minor premise** — together reach a **logical conclusion**.
- For example, the major premise "Every A is B" could be followed by the minor premise, "This C is A." Those statements would lead to the conclusion "This C is B."
- Syllogisms are considered a good way to test deductive reasoning to make sure the argument is valid.

Hypothesis (must be correct)

$$A = B$$



Examination

$$B = C$$



Logical conclusion (true)

$$A = C$$

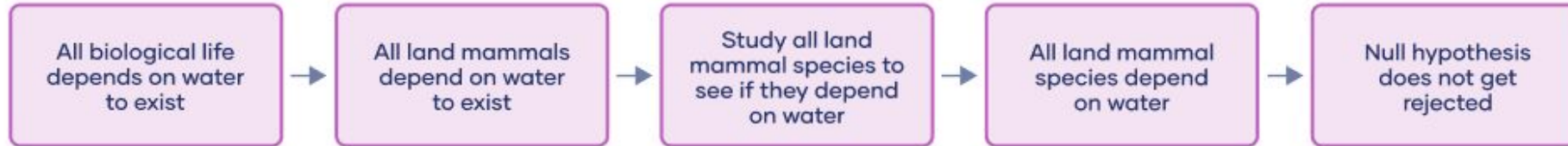
Deductive Reasoning



Deductive reasoning

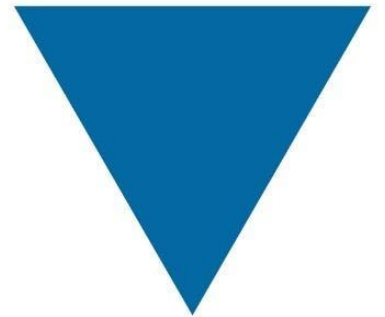


Example



DEDUCTIVE REASONING

BROAD



SPECIFIC

Examination



Data



Conclusion
(logical but may not be true)

Inductive reasoning involves following questions:

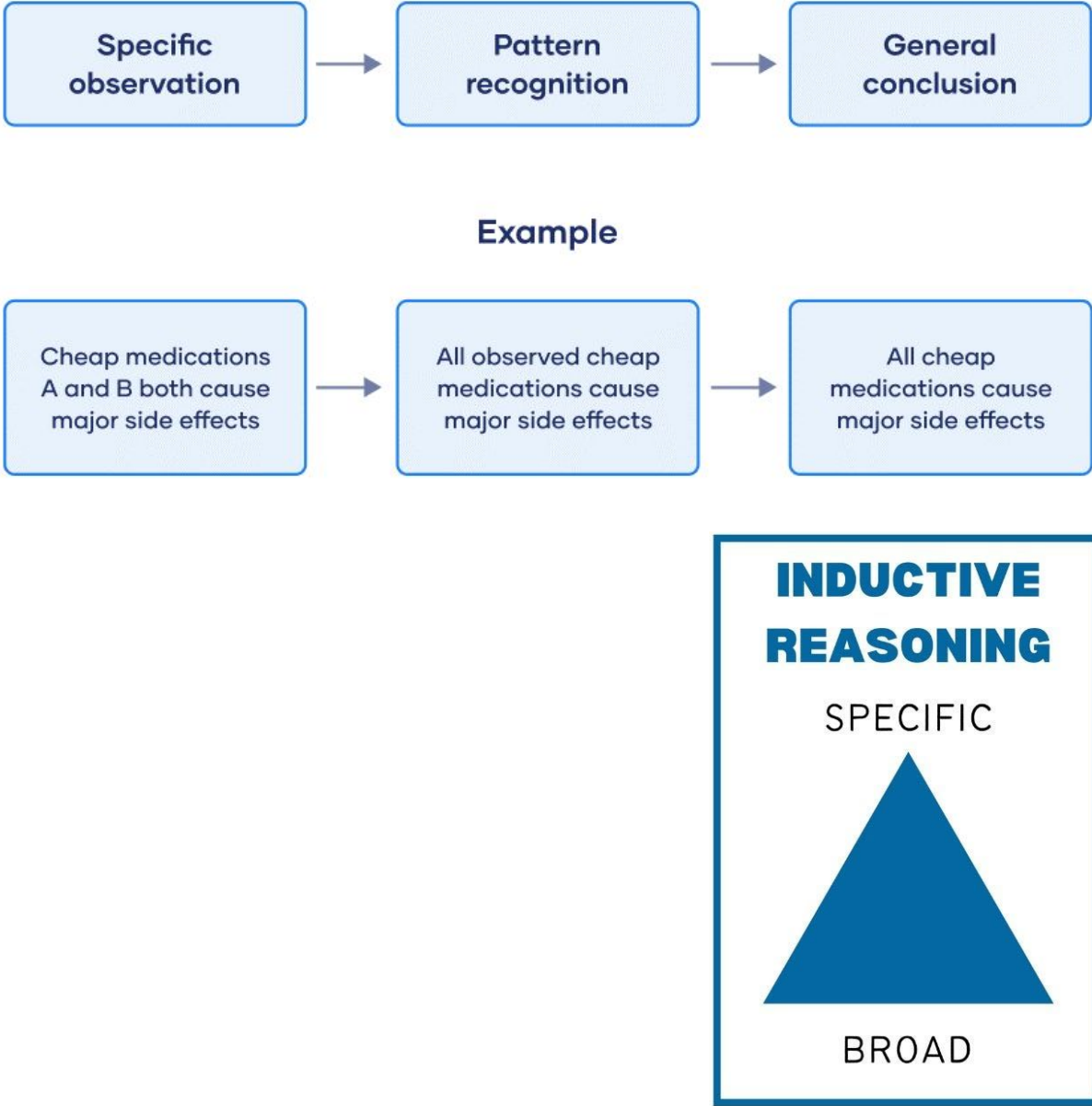
1. What have you observed?
2. What evidence is available?
3. What can you conclude from the evidence?
4. Is that conclusion logical?

Example: Cobra is venomous

Cobra is a snake

All snakes are venomous

Inductive Reasoning



INDUCTIVE REASONING

Inductive reasoning is the process of deriving general principles or conclusions based on specific observations or examples.

DEFINITION

Inductive reasoning involves moving from specific instances or observations to broader generalizations. It relies on collecting evidence, patterns, or trends from the specific cases and using them to make probabilistic or plausible conclusions about a larger set of cases or the entire population.

EXAMPLE

"The sun has risen every morning throughout history; therefore, the sun will rise tomorrow."

This example showcases inductive reasoning by extrapolating from past observations of the sun rising to predict that it will continue to rise in the future.

Incomplete observation



Likeliest explanation



Conclusion/Argument
(may not be true)

- useful for forming hypotheses to be tested

- used by doctors who make a diagnosis based on test results

- used by jurors who make decisions based on the evidence presented to them

Abductive Reasoning

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The bus is late.

I heard about a motorbike crash on the radio.

The crash has probably held the bus up.

That man looks tired and he's wearing gym clothes.

He's probably been to the gym.

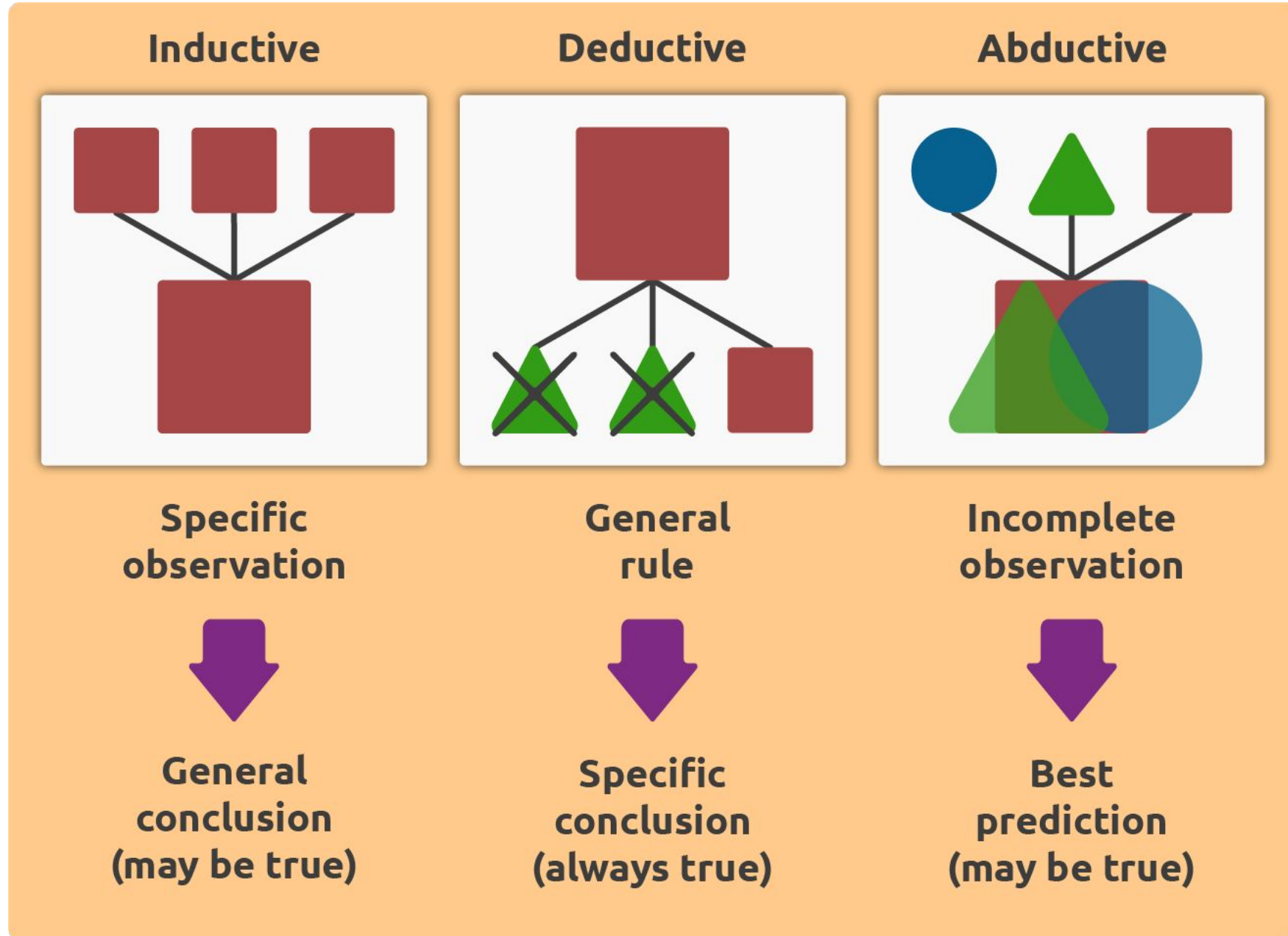
Cleaning the wound seems to prevent infection.

Infectious materials might be carried in dirt.

In all of these cases we are starting with a fact (or a conclusion) and working backwards to try to find an explanation that fits best.

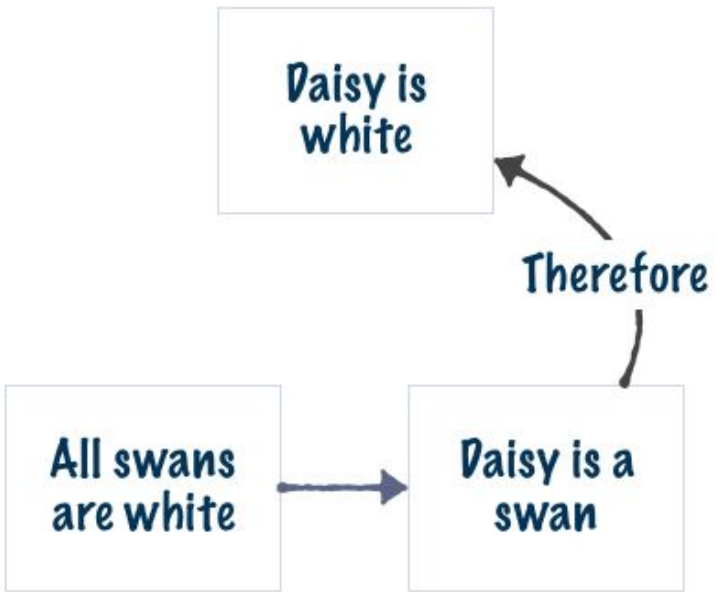
Hence: "*inference to the best explanation.*"

Comparison

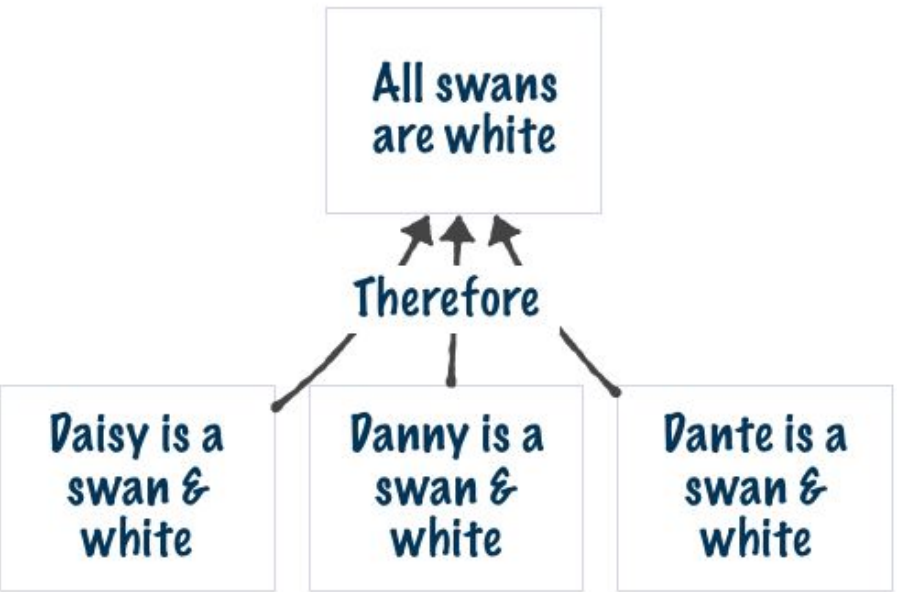


Comparison

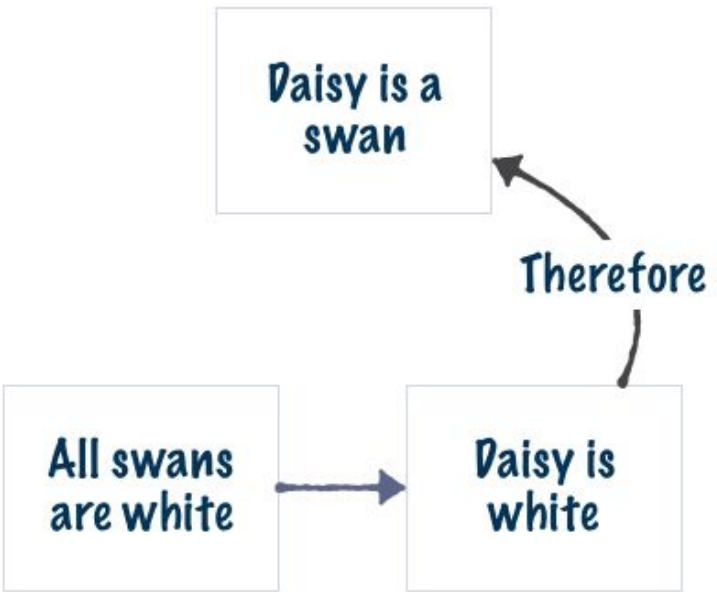
DEDUCTION



INDUCTION



ABDUCTION



You need to answer this

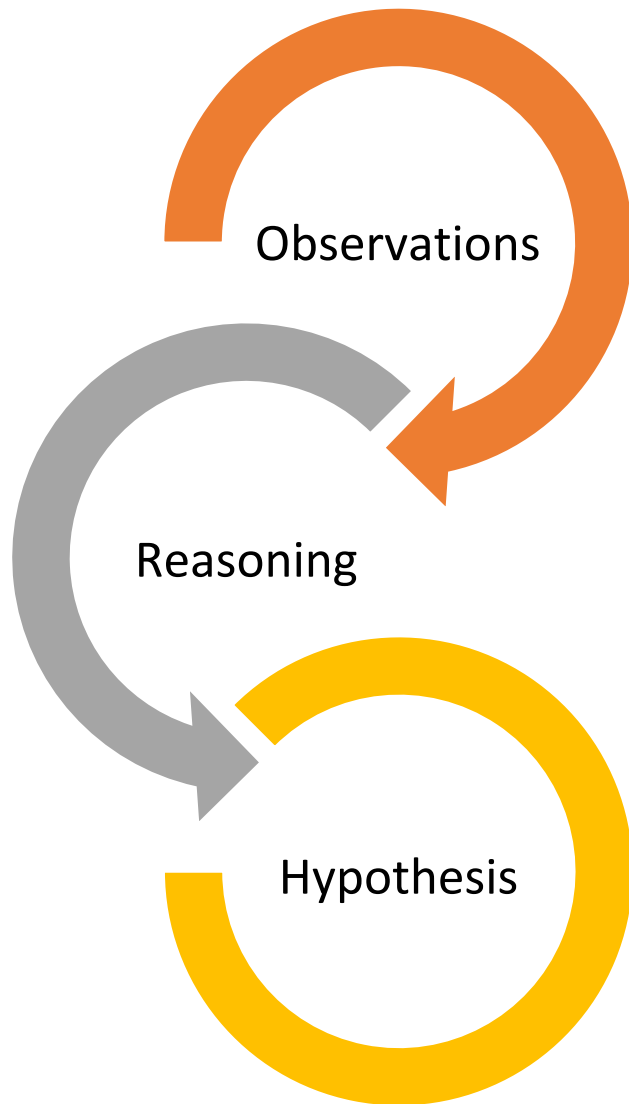


Statement 1: Biology is boring

Statement 2: Bioengineering contains Biology

Conclusion: Bioengineering is boring

What kind of reasoning is this? Think it over!



Biology & Engineering Approaches:

- Observations; deductive, inductive and abductive reasoning;
- Hypothesis testing; reductionism; systems biology; modeling;
- Tinkering with biological systems; cause & effect; translational research; attributes of a Bioengineer